

Sensor-Integrated Head Design for Robust Perception in Humanoid Robots

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The AMBER Lab at Caltech conducts research on locomotion, control, and human–robot interaction, aiming to build robots that can safely operate in human environments. Recently, the lab has been working with the Unitree Humanoid G1, a humanoid designed for research in robotics and human–robot interaction. Ensuring safety in humanoid operation requires robust perception, as the robot must reliably sense and respond to its environment. This project focuses on enhancing perception of the G1 robot through the development of a new head design that integrates multiple visual sensors. The redesigned head integrates four RGB-D cameras, providing a comprehensive field of view for safer navigation and interaction. Early results show smooth integration of the cameras with ongoing progress in visualization. Future work will refine the head’s mechanical design to further improve perception, robustness, and safety.